

Introduction to Web Design

Lecture Objectives

- By the end of this lecture, students should be able to:
1. Understand what web design is and why it is important
 2. Distinguish between websites and web applications
 3. Understand the basic components of a website
 4. Know the roles of HTML, CSS, and JavaScript
 5. Understand how websites work (client–server model)

Introduction to Web Design

1- What is Web Design?

Web design is the process of planning, creating, and maintaining websites. It includes:

1. Structure (HTML)
2. Appearance (CSS)
3. Behavior/interaction (JavaScript)
4. A good web design should be:
5. Easy to use (Usability)
6. Visually clear
7. Fast and responsive
8. Accessible on different devices (PC, mobile, tablet)

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BAD



VS

GOOD



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1. Your website is often the first impression of your brand, products, or services.
2. It is important to avoid common design mistakes that annoy users.
3. Good website design focuses on user experience.
4. A good design makes information clear, visually attractive, and easy to navigate.
5. Bad website design makes it hard for users to reach their goals.
6. Poor design also makes navigation and understanding content difficult.

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2- Website vs Web Application

A- **Website:** Mostly provides information (e.g., news site, company site).

B- **Web Application:** Allows user interaction and processing (e.g., email, online banking).

3. Main Components of a Website

1. Frontend (Client-side)

1. What the user sees in the browser
2. Uses: HTML, CSS, JavaScript

2. Backend (Server-side)

1. Handles data, logic, authentication
2. Uses: PHP, Python, Node.js, Java, etc.

3. Database

1. Stores data (users, posts, products, etc.)

FRONT-END



Web App

Mobile App

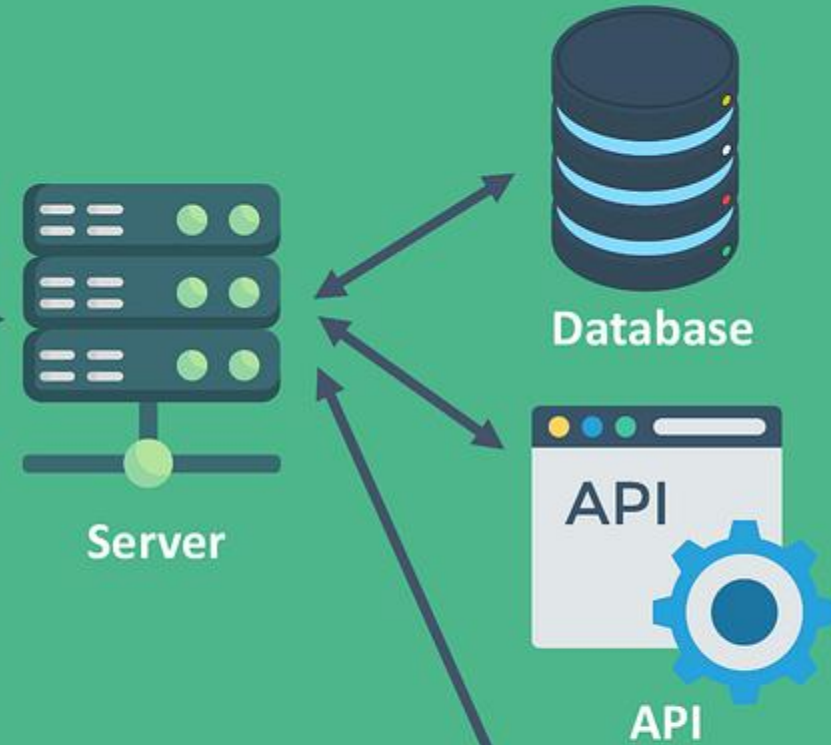
HTML

CSS

JS



BACK-END



Server

Database

API



Java



PHP

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4- Core Web Technologies

1. HTML (**HyperText Markup Language**) → Structure of the page
2. CSS (**Cascading Style Sheets**) → Design and layout
3. JavaScript → **Interaction and behavior**

Example:

HTML = skeleton

CSS = clothes and style

JavaScript = movement and actions

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7- introduction to Responsive Design

With more users browsing on mobile, having a non-responsive website is a major mistake. A responsive site adapts to all screen sizes and provides a smooth experience, while a non-responsive site forces users to zoom and scroll, causing frustration and higher bounce rates.

A- Responsive design means:

- 1- The website adapts to different screen sizes.
- 2- Works well on mobile, tablet, and desktop.

B- Non-responsive design: A non-responsive website requires users to pinch, zoom, and scroll to navigate; this can lead to significant frustration and often high bounce rates.

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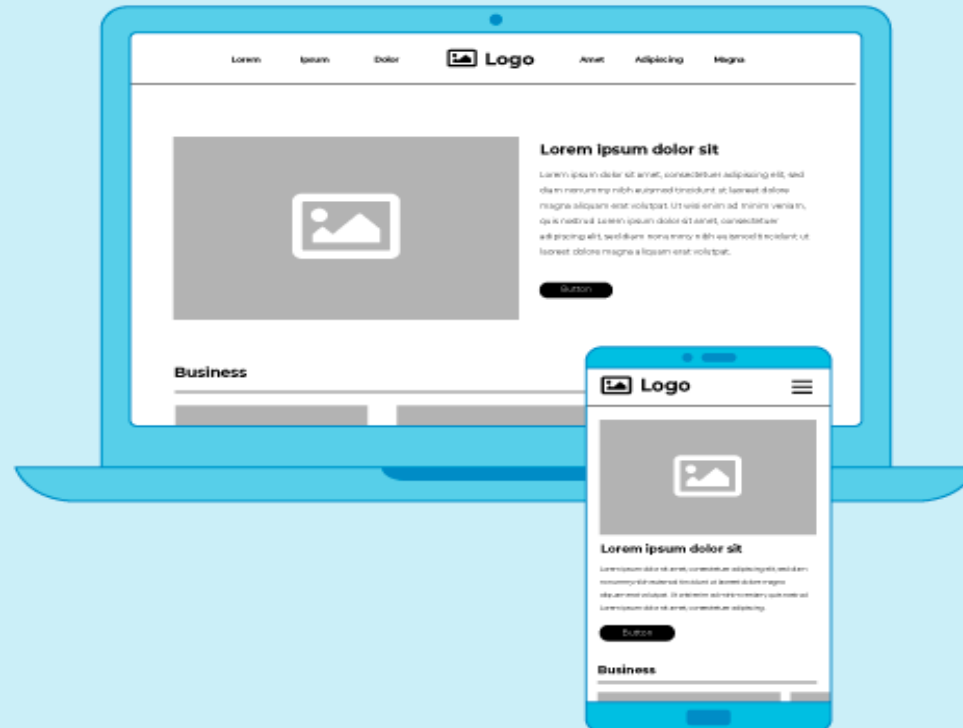
5- How the Web Works

1. User types a website address in the browser
2. Browser sends a request to the server
3. Server sends back HTML/CSS/JS files
4. Browser displays the page to the user

6- Tools for Web Design

1. Web browser (Chrome, Firefox, Edge)
2. Code editor (VS Code, Notepad++, etc.)
3. File system (folders for project files)

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Notes about Responsive Non-responsive design

1- Poor Navigation: Easy-to-use navigation is essential if users are to find the information they're seeking and perform the tasks they want to achieve with your website. **Examples of poor navigation can include over-complicated menus with excessive numbers of sub-categories, or hidden menus with vague labels.**

2- Cluttered Layouts :Cluttered layouts overload users with **too much content and visual noise, making it hard to focus.** A clean, minimalist design with good organization, clear hierarchy, and enough white space helps highlight important information and guide users smoothly through the content.

3- Stylistic Inconsistency: Stylistic inconsistency makes a website look unprofessional and confusing. Using consistent fonts, colors, and design elements across all pages creates a unified, trustworthy look and improves user recognition and ease of use.

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4- Slow Loading Speed: Users expect websites to load quickly and will leave if they have to wait too long. **Slow loading speeds increase bounce rates and harm search engine rankings, so improving performance with caching, optimized code, smaller files, and fewer ads is essential.**

5- Lack of clear CTAs: Without clear **CTAs (Call to Actions)**, users may not know what to do next, leading to missed conversions. **Strategically placed, easy-to-find CTAs with clear action words help guide users and improve user journeys.**

Example: Shop Now, Contact Us, Get Started.